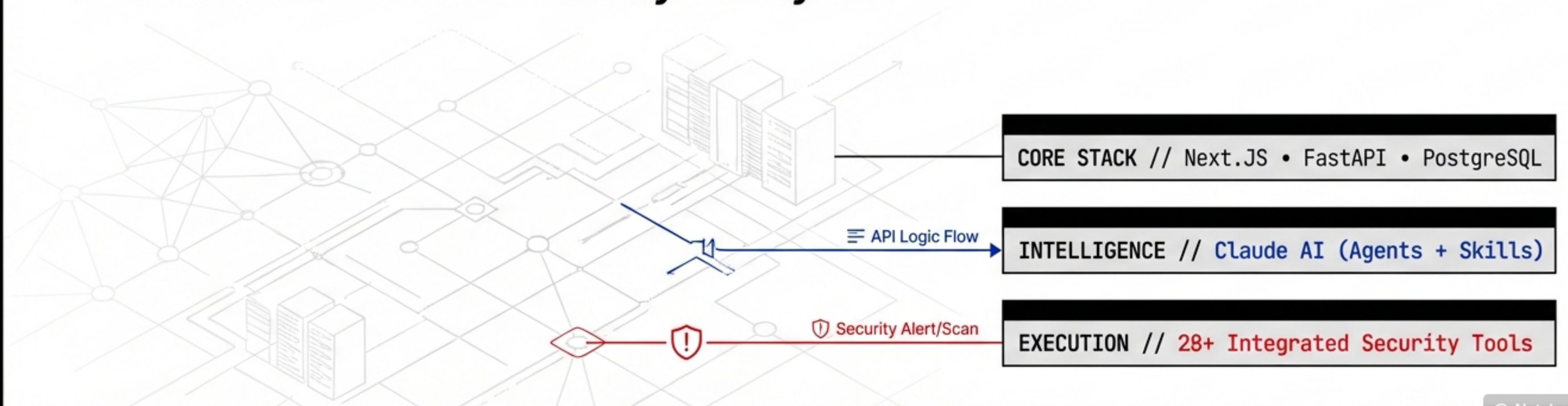
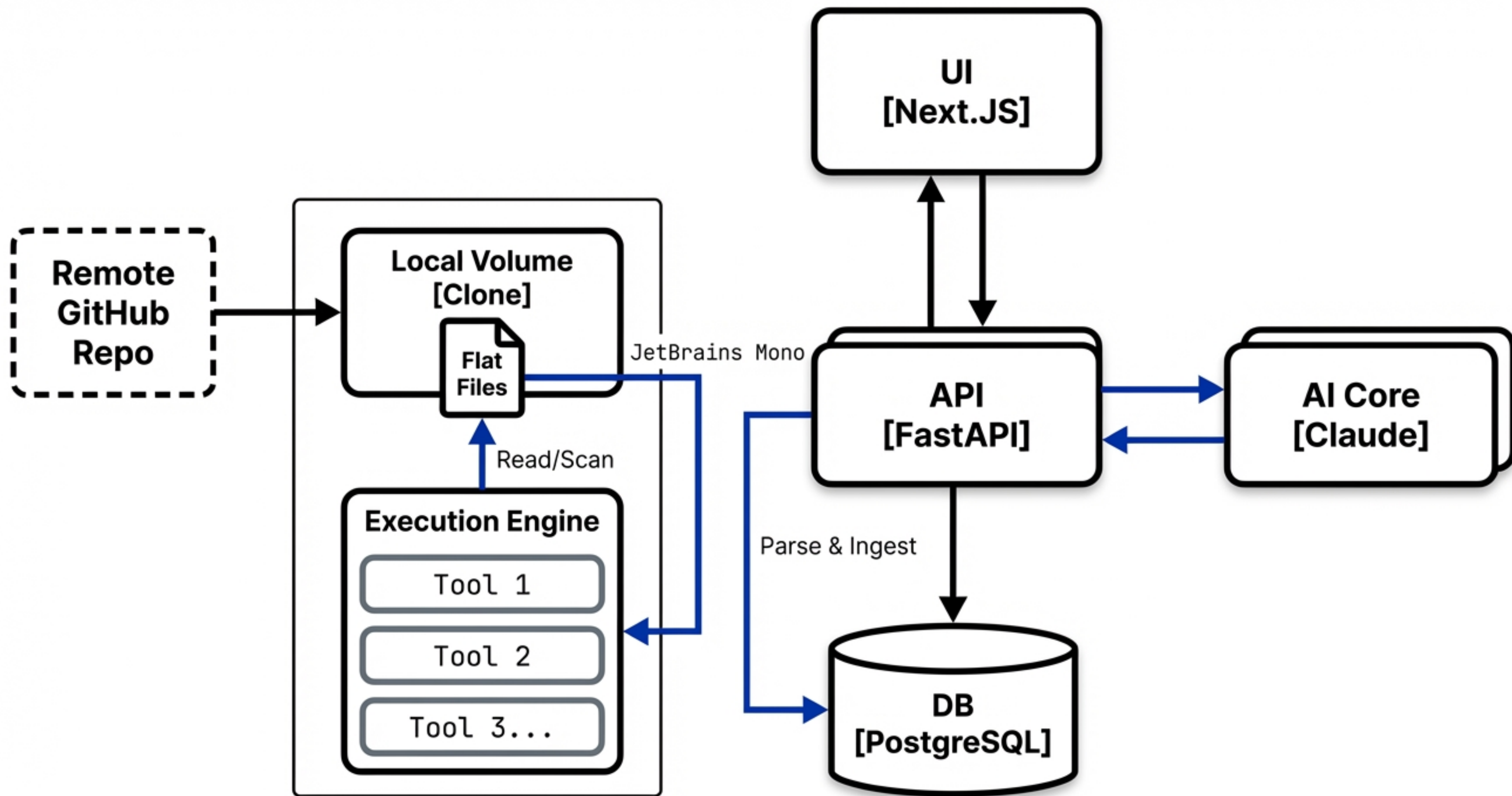


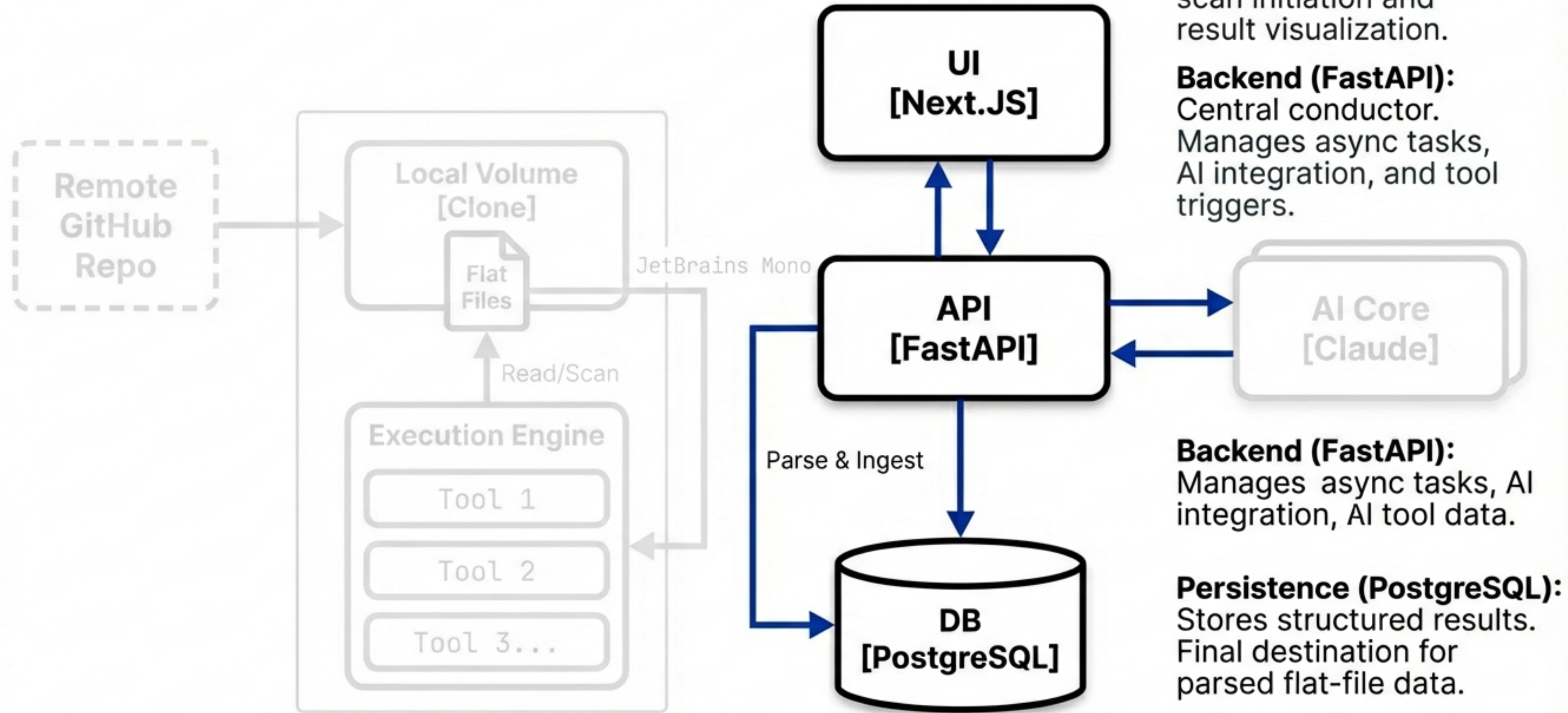
Architecture Overview: AI-Augmented Security Security Scanning

Orchestrating Next.JS, FastAPI, and Claude
for Automated Vulnerability Analysis

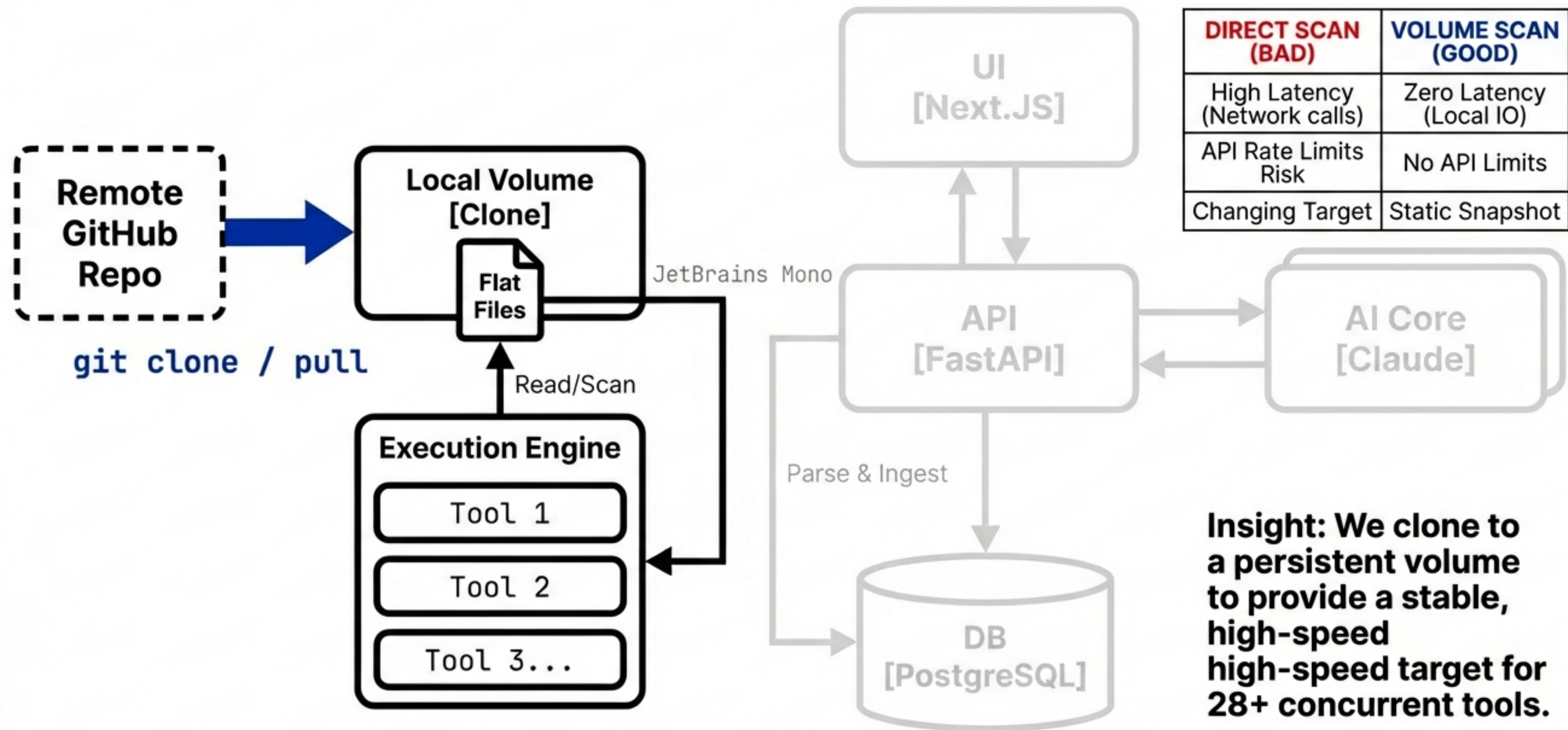




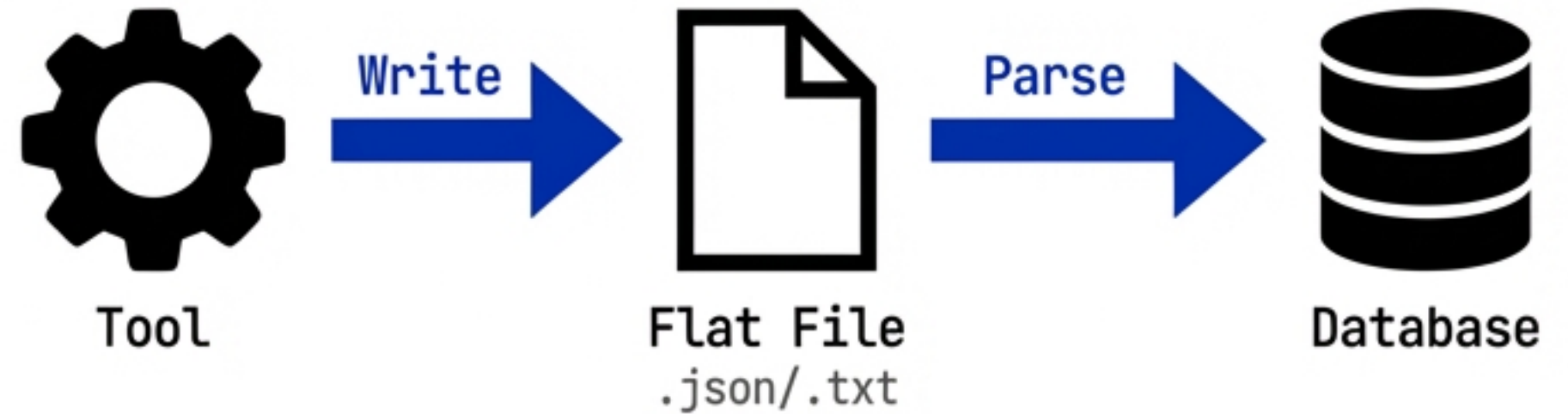
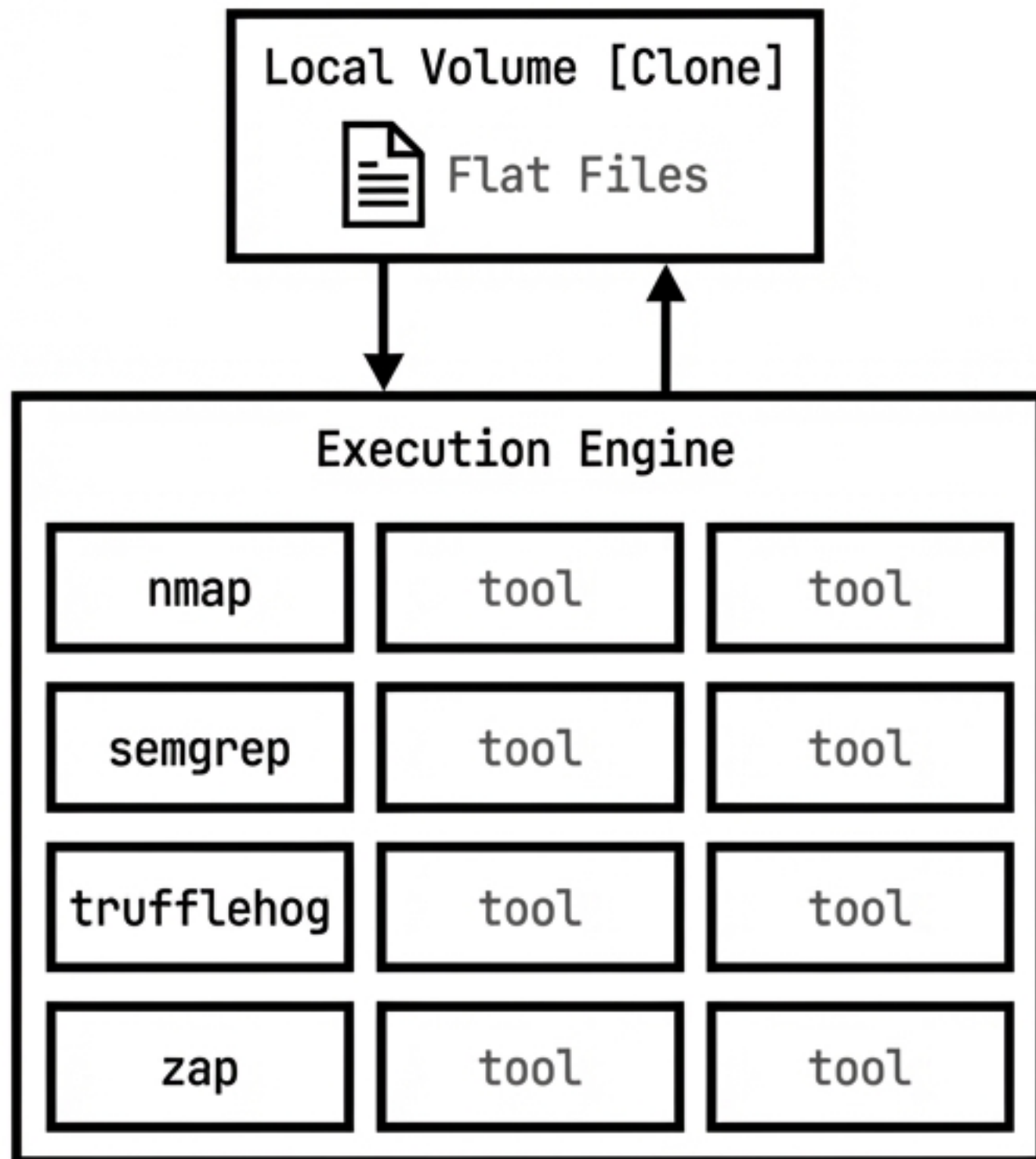
The Application Layer: Orchestration & State



Target Isolation: The Local Volume Strategy



The Execution Engine

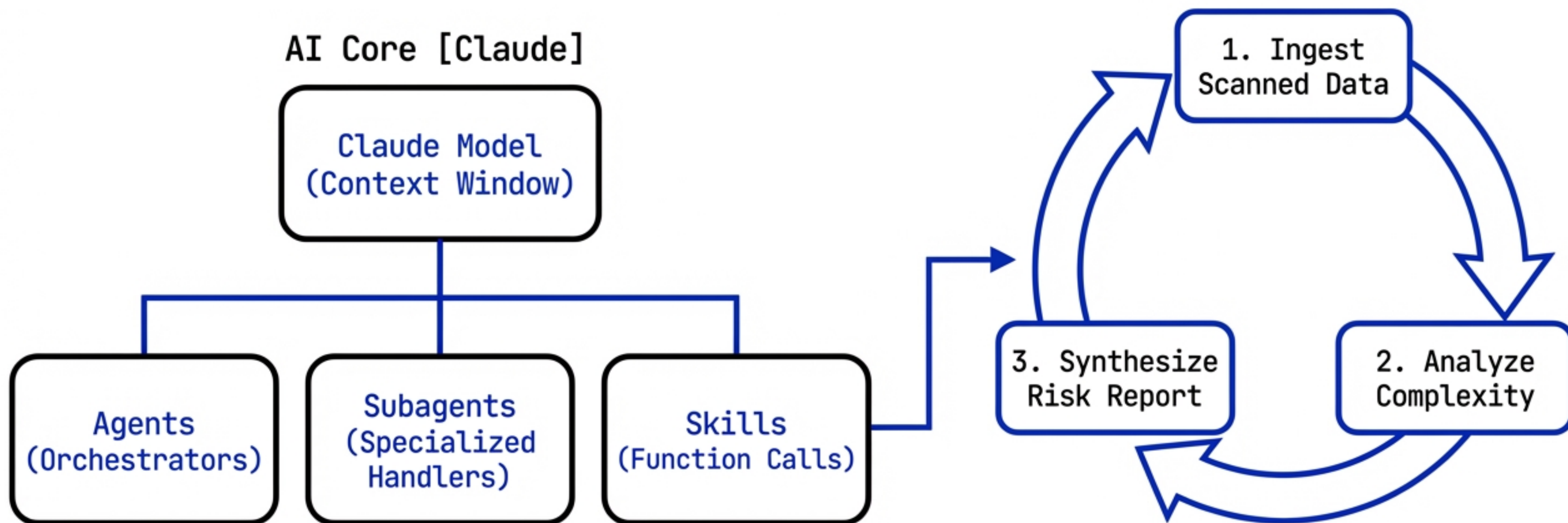


THE WRITE PATTERN:

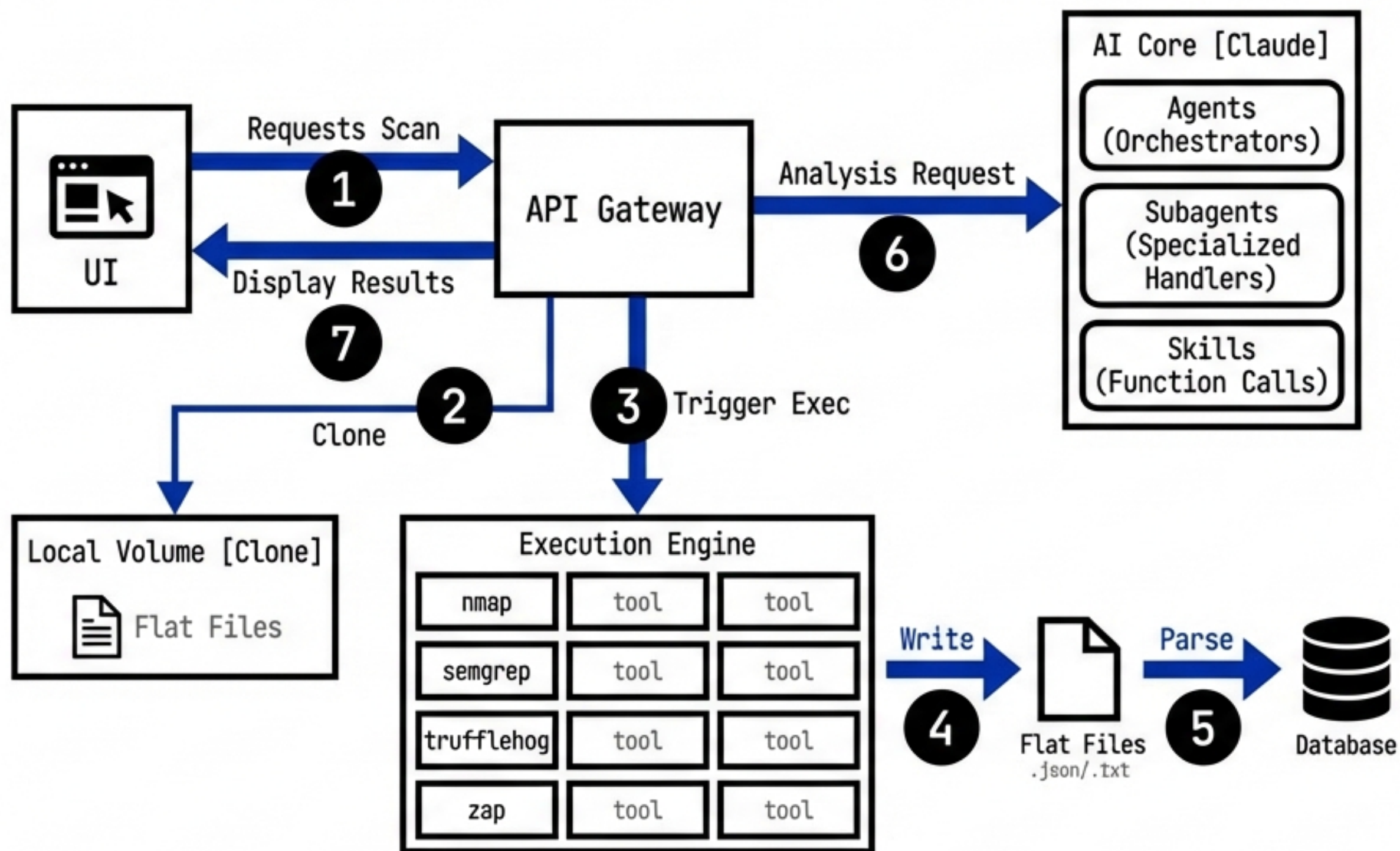
1. Tools execute against Local Clone.
2. Output is captured as raw Flat Files in the Volume.
3. Post-process parses files -> Inserts to Postgres.

Why? Decouples tool execution from DB schema strictness.

The Cognitive Core: Adaptive AI Modeling



End-to-End Data Flow



1. **Initiation:** User requests scan.
2. **Cloning:** Repo synced to local volume.
3. **Execution:** 28+ tools run against volume.
4. **Output:** Raw results written to files.
5. **Ingestion:** Data parsed into Postgres.
6. **Synthesis:** Claude analyzes findings.
7. **Display:** Results pushed to UI.

Key Takeaways

SCALABLE ARCHITECTURE

Decoupling tool execution from database schemas allows for the easy addition of new tools without breaking the data model. Flat file buffering ensures resilience.

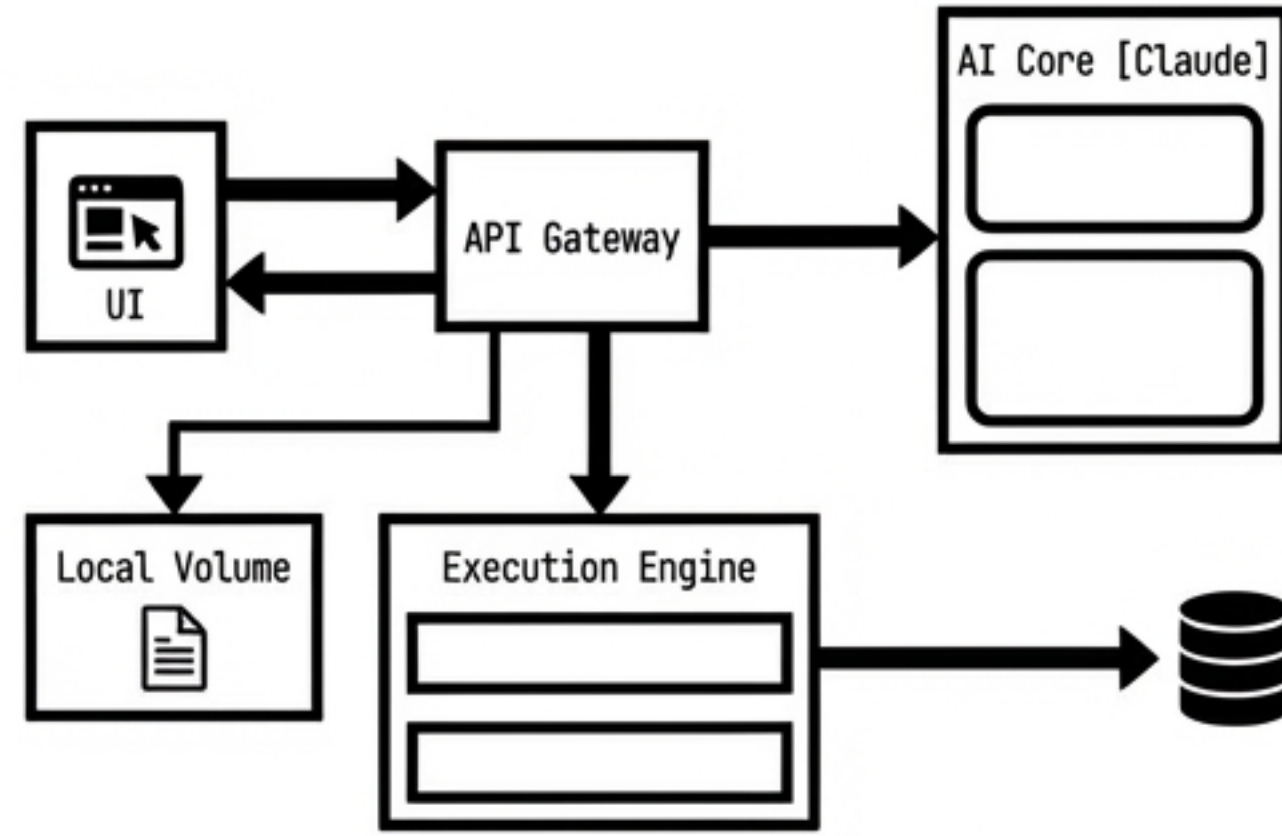
INTELLIGENT ANALYSIS

Claude is not just a chatbot here; it acts as an orchestrated analyst, using specialized agents and skills to triage vulnerability data based on complexity.

SECURE & FAST

Scanning against a local cloned volume removes network latency and protects the production repository from API rate limits and excessive traffic.





Architecture Review Complete.

Questions & Discussion